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Domain adaptation based automatic identification method of vortex induced vibration of long-span bridges without prior information

Chunfeng Wan [a], Jiale Hou [b,*], Guangcai Zhang [a], Shuai Gao [a], Youliang Ding [a], Sugong Cao [c], Hao Hu [c], Songtao Xue [d,e,**]

[a] *School of Civil Engineering, Southeast University, Nanjing, China*
[b] *Department of Civil Engineering, Tsinghua University, Beijing, China*
[c] *Key Laboratory of Road and Bridge Detection and Maintenance Technology of Zhejiang Province, Hangzhou 311305, China*
[d] *Department of Disaster Mitigation for Structures, Tongji University, Shanghai, China*
[e] *Department of Architecture, Tohoku Institute of Technology, Sendai, Japan*

A R T I C L E I N F O

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A B S T R A C T

Machine learning algorithms can sensitively capture the characteristics of vortex induced vibration (VIV) of the girder in long span bridge from the extensive historical data accumulated by structural health monitoring (SHM) system over several years. These algorithms have gradually become a promising method of VIV identification. However, the algorithms proposed by previous researchers require historical VIV data to select the threshold or parameters to identify VIV. Most long-span bridges have not recorded a significant amount of VIV data since VIV is rare, or the bridge were not equipped with SHM system before. This study proposes an adaptive VIV identification method based on domain adaptation methods, which can identify VIV in real-time or in historical monitoring datasets of the target bridge without prior VIV information or parameter settings. The strong generalization ability of the proposed method is verified on the SHM dataset of two long-span suspension bridges in China. It is found that the VIV recognition accuracy of the balanced distribution adaptation (BDA) based VIV identification method is higher than that of other algorithms. In this study, the BDA based algorithm is also applied to the 8 months monitoring datasets of a long span bridge and successfully identifies more than 20 VIV events of the main girder, which has shown the stability and accuracy of the proposed algorithm.

1. Introduction

With the increase of bridge span, the structural vibration characteristics of the bridge itself change significantly, such as damping reduction, stiffness reduction, low-frequency dominant and so on. For long-span suspension bridge, the wind-induced effects become the major cause of disasters. Scholars have discussed many wind-induced phenomena of long-span suspension bridges or their components through experiments and numerical model simulation. Among them, vortex induced vibration (VIV) is a special wind-induced phenomenon, and so far, it has not been studied clearly. This phenomenon has attracted extensive attention (Chen et al., 2015; Ge et al., 2022; Huang et al., 2013; Han et al., 2021; Li et al., 2011a), and has been observed occasionally in several long-span bridges all over the world, such as Volgograd continuous bridge with steel box girder in Russia (Weber and Maślanka, 2012), Yi Sun-sin suspension bridge in Korea (Hwang et al., 2015), Xihoumen Bridge (Li et al., 2011a) and Humen Bridge in China (Zhang et al., 2021a; Zhao et al., 2022), etc.

VIV is a complex aeroelastic wind induced response caused by nonlinear fluid and structure interaction (Yu et al., 2022). When the uniform airflow passes through the bluff body, such as a bridge deck, it will alternately produce periodic vortexes shed from the structural surface on both sides of the bluff body. This periodic vortex will produce pulsating pressure on the bluff body surface, which will have a periodic force on the structure. When the bridge deck is excited by the vortex whose shedding frequency is closed to the natural frequencies of the deck, the resonance self-excited vibration of the bridge deck will occur. Once the bridge deck is set to resonance vibration, the vortex shedding frequency of the airflow will be locked on the vibration frequency of the deck, so that the VIV can continue within a certain wind speed range.

* Corresponding author.
** Corresponding author. Department of Disaster Mitigation for Structures, Tongji University, Shanghai, China.
E-mail addresses: hjl23@mails.tsinghua.edu.cn (J. Hou), xue@tongji.edu.cn (S. Xue).

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Due to the fluid-structure interaction, VIV is highly nonlinear dynamic system and difficult to predict.

Some researchers have conducted field measurements to investigate the dynamic characteristics of VIV and its correlation with wind field environment. Smith (1980) measured the wind induced dynamic response of the 235 m span cable stayed bridge over the River Wye using automatic recording equipment. The large response of the bridge occurs in the range of 7–8 m/s wind speed, and the wind direction is approximately perpendicular to the axis of the bridge. The results are similar to the predicted results of the wind tunnel test, but the predicted response amplitude is lower than field-measured results. Fujino et al. (Fujino and Yoshida, 2002) investigated the VIV event of the Trans-Tokyo Bay Crossing Bridge and found that the results of field test and wind tunnel test are consistent in terms of VIV amplitude and wind speed range of the first vertical vibration mode of the bridge, and the first mode can only be observed when the wind direction is within $\pm 20^{\circ}$ of the bridge transverse axis. Since the vertical displacement amplitude has been 50 cm during the VIV, tuned mass damper (TMD) is used to enhance the aerodynamic control of the first two modes. Frandsen (2001) studied the measured wind pressure and acceleration data of VIV events of the Great Belt East Bridge and calculated the Strouhal number of the bridge is about 0.08–0.15. The vertical displacement amplitudes up to 30 cm occurred at wind velocity about 4–12 m/s, while the wind direction is nearly perpendicular to bridge axis, with low turbulence intensity. Similar observations have been reported by Li et al. (2011a). Kim et al. (2017) studied the relationship between the amplitude of VIV and identified damping ratio based on the measured data of Jindo bridge, and designed a multiple tuned mass damper (MTMD) to significantly reduce the amplitude of VIV.

Bridge structure health monitoring (SHM) system has gradually become an important method to monitor the static and dynamic responses in real time (Azimi et al., 2020; Bao et al., 2019; Gordan et al., 2022), and has been widely installed on important long-span suspension bridge structures (Annamdas et al., 2017; Comisu et al., 2017). The datasets collected by bridge SHM system contain a huge amount of data of bridge response, including the VIV data. Most of the field-measured responses are environmental vibration responses caused by wind and vehicle load, and only a few are VIV responses, which is difficult to identify manually (Dan and Li, 2022; Huang, 2013). In addition, the bridge SHM system shall be able to identify VIV in real time and then report to the traffic management department and public, to reduce the psychological panic of vehicle drivers on the bridge deck caused by large vibration (Cao et al., 2020). Therefore, the automatic identification of VIV response in real time and in large-scale historical monitoring dataset are the important practical need to study VIV.

Consequently, studies have been conducted with the goal of identifying VIV in real time or in massive monitoring datasets. In particular, with the rapid development of machine learning (ML), deep learning (DL), the data-driven VIV identification method can be easily applied to the massive data accumulated by the SHM system. Hua et al. (2018) employed the autoencoder to automatically extract the characteristics of SHM data, and set the threshold to determine whether it is VIV. Huang et al. (2019) used the random decrement method to preprocess the monitoring data, in which the variation coefficient threshold of the processed data peak was used to detect VIV. Zhao (Zhao et al., 2021) et al. studied the monitoring data of a VIV event of a suspension bridge, and proposed an early warning framework of VIV by setting thresholds for wind parameters and vibration parameters. Li et al. (2017) proposed an unsupervised ML-based framework for automatically classifying VIVs from the acceleration signal of a suspension bridge from 2010 to 2014 and VIV events with six dominant frequencies are identified. Arul et al. (2020a) identified the VIV from the historical data of the Burj Khalifa from 2010 to 2014. The acceleration data features are extracted firstly, and then the dimension of data features is reduced by principal component analysis (PCA). Finally, the VIV part is successfully identified by cluster analysis of the first two principal components. Lim et al. (2022a) proposed a three-stage strategy, containing semi-supervised labeling, DNN model training, and parameter optimization, to identify VIV in massive historical dataset.

Some of the above methods require historical VIV data identification to determine the threshold or parameter for VIV identification (Hua et al., 2018; Huang et al., 2019; Zhao et al., 2021; Lim et al., 2022a), however, many long-span bridges in China have no VIV data or the data has not been recorded, so an appropriate threshold or parameter for such bridges is difficult to determine. Other clustering methods can only identify the VIV in the massive historical data accumulated for several years in SHM system (Li et al., 2017; Arul et al., 2020a), but it is difficult to identify the VIV in real time. Therefore, it is crucial to develop an adaptive VIV recognition algorithm that does not require VIV historical data of the target bridge and can identify VIV in real-time or in historical monitoring datasets.

Transfer learning (Lu et al., 2015; Zhuang et al., 2021) captures the similarity between data, tasks, or models to apply the models and knowledge learned in the lab domain (source domain) to the new domain (target domain), which has attracted the attention of many researchers in civil engineering (Ardani et al., 2023; Yang et al., 2023). Zhang et al. (2021b) used convolutional long short-term memory network and Joint Distribution Adaptation (JDA) method to transfer the structural damage identification algorithm of one sensor of a structure to the multi-sensor of another structure, and achieved the damage imaging visualization. Li et al. (2021) proposed a deep learning method enhanced by transfer learning to predict the long-term monitoring displacement data of the dam, which is obviously better than long short-term memory network, support vector regression and other methods.

This article is a continuation of the previous work which is also inspired by transfer learning (Hou et al., 2023). In the previous research, the author used Transfer Component Analysis (TCA) to automatically recognize VIV events of the girder in long span bridge. This article developed VIV identification method based on different domain adaptation methods including TCA, JDA and Balanced Distribution Adaptation (BDA). The advantages of the proposed method are listed here:

- The historical VIV data of the target bridge is not needed;
- The parameter settings and selection are not required;
- This method can be used for both real-time VIV determination and analysis of VIV within historical datasets.

However, in the proposed method, manually labeled monitoring data from another long-span bridge containing VIV events is required. It is used to be aligned with the monitoring data of the target bridges using domain adaptation methods, which means the unknown VIV information of the target bridge is replaced with the VIV information from other bridges. Consequently, the VIV identification algorithm for the target bridge can be trained using the monitoring dataset of the target bridge, alongside manually labeled monitoring data obtained from another bridge. This article uses two large span bridge health monitoring datasets to validate the proposed domain adaptive based VIV recognition algorithm. It is ultimately found that the BDA based VIV recognition accuracy is the highest, and the algorithm is applied on a long-term monitoring dataset on one of the bridges, ultimately identifying more than 20 VIV events. This paper is organized as follows. In Section 2, monitoring datasets used in this article obtained from two long span suspension bridges is analyzed. Then, four domain adaptation methods are introduced, and the framework of the proposed method is established in Section 3. Case studies on the monitoring datasets of the two bridges are conducted in Section 4 and the algorithm application on the long term monitoring dataset of a long span bridge is conducted in Section 5. In the end, conclusions are remarked in Section 6.

2. Dataset

This study validated the proposed algorithm using monitoring datasets of two large span bridges with different spans and located at different places in China. As mentioned earlier, this algorithm does not require the target bridge to have VIV monitoring data, only NV (normal vibration) and VIV data in the source domain dataset from another bridge. In order to verify the effectiveness of the proposed algorithm in identifying VIV events, the monitoring datasets of both bridges used in this article contain both NV and VIV data.

2.1. Bridge A

The bridge A investigated in this study is a long span suspension bridge, which is located on the Changjiang River in Wuhan linking Hanyang and Wuchang. The main bridge is the world's first three-tower four-span steel-girder concrete-deck suspension bridge with a span arrangement of (225 + 2 × 850 + 225) m. The elevation of the intersection between the cable centerline and the top of the middle tower is +162.5 m, and the elevation of the intersection between the cable centerline and the top of the side tower is +144.5 m. The steel-box girder of the bridge is comprised of 8 lanes measuring 36 m in width and supported by 264 cables. The layout, dimensions of the bridge appear in Fig. 1.

Since the opening of the bridge, a SHM system has been installed and operated to monitor the in-service conditions. Fifty-eight sensors were installed on the bridge, including thermometers, anemometers, inclinometers, displacement meters and accelerometers sensors. Fig. 1 shows the layout of partial sensors that used in this study. The hydrostatic leveling instrument is used to measure the vertical deflection of the girder, which is installed on Sections B to X. A longitudinal acceleration sensor is set on the top of the middle tower and transverse and vertical acceleration sensors of the girder are set on bottom center of Section I, K, L and Q. In each of Sections L and N, transverse and vertical acceleration sensors are set on cable to measure the vibration of the main cable, respectively. The sampling frequency of the acceleration sensor is 20 Hz, and the sampling frequency of all other types of sensors is 0.25 Hz. For more information about the bridge and the SHM system, please refer to this paper (Zhao et al., 2021).

The bridge A, which was opened to public in December 2014, exhibited unexpected twice VIVs for about 2 h in April 26, 2020. As is shown in Fig. 2 (a), the wind direction remains stable at 200° from 7:50 to 17:25, and the wind speed, ranged from 5.732 to 8.546 m/s during this period, is higher than other time. And it can be seen from wind rose diagram that the stable wind direction is nearly perpendicular to the axial direction of the bridge.

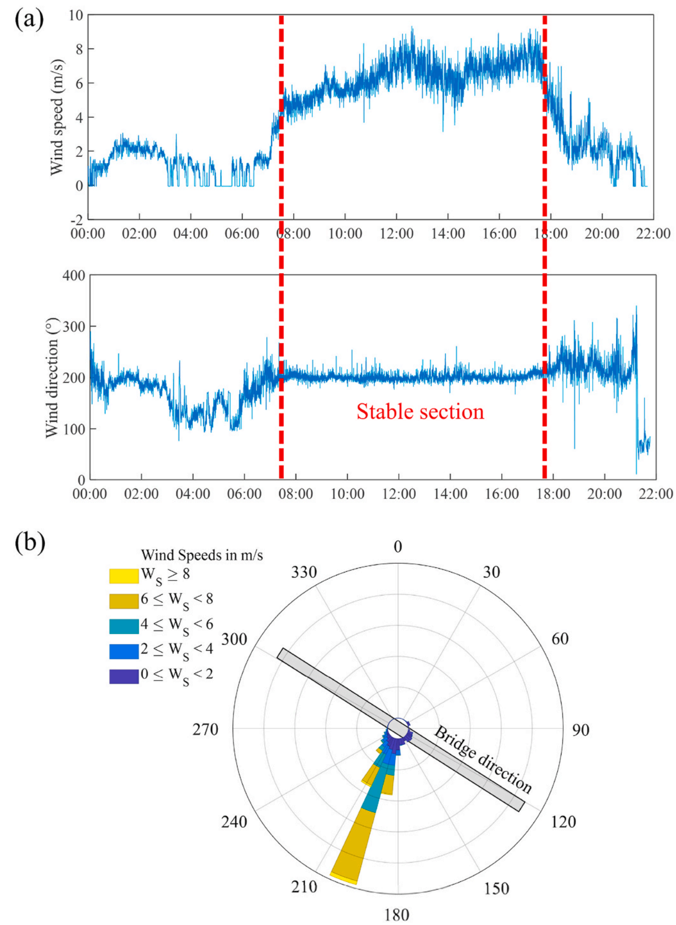


Fig. 2. Wind environment of bridge VIV phenomenon: (a) time history data of wind speed and direction; and (b) wind rose diagram.

The acceleration data of all section after 17:00 is missing for some unknown reasons. The vertical acceleration of Section Q (mid span section of main span in Wuchang direction) and Section I (mid span section of main span in Hanyang direction) from 0:00 to 17:00 is shown in Fig. 3. It can be seen from Section Q data that the acceleration amplitude suddenly increased for the first time from 12:00 and lasted for about 1 h, which is named as VIV event 1. Then the acceleration

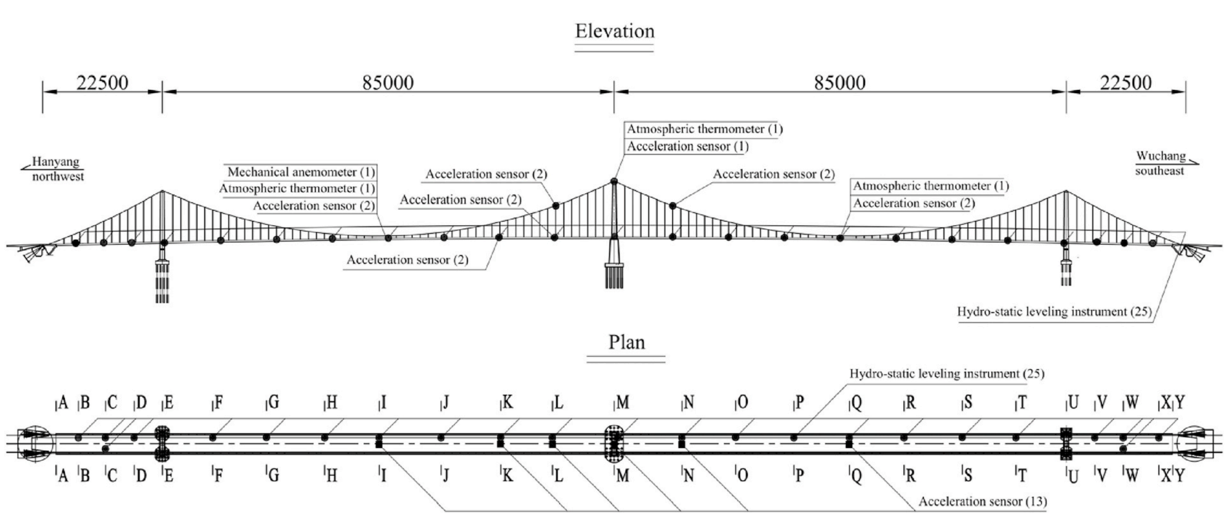


Fig. 1. Partial arrangement of monitoring sensors of bridge (unit: meters).

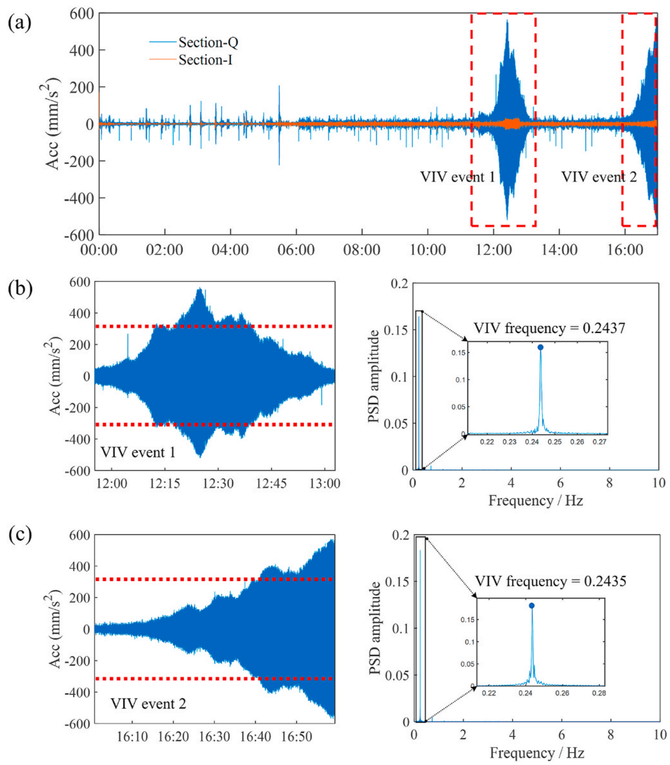


Fig. 3. The vertical acceleration data in April 26, 2020: (a) acceleration data of Section Q and Section I; (b)VIV event 1 and PSD; (c)VIV event 2 and PSD.

returned to the normal amplitude, but at 16:10, the acceleration suddenly increased again, which is named as VIV event 2. As shown in Fig. 3, Section Q maximum acceleration of the two VIV events exceeded the blue warning value of 315 mm/s^2 (displayed as dotted line), as suggested by the recently issued Chinese standard for bridge health monitoring (Moreu et al., 2018). The time of two VIV events coincides with the time of wind direction stability stage and there may be a certain causal relationship between them. The dominant frequencies of the two observed VIV events are very close, 0.2437 and 0.2435 Hz respectively. The fundamental frequencies of the bridge A are identified through finite element model by other researchers and listed in Table 1. It can be concluded that the frequency of the VIV event in bridge A corresponds to the 8th fundamental frequency 0.248 Hz, which means the mode shape of this VIV event is the fourth order vertical antisymmetric bending. Besides, the acceleration amplitude of Section I increases slightly during the VIV event, and the maximum amplitude obviously does not reach the warning threshold specified in the standard. Therefore, it can be concluded that two VIV events mainly occur on the main span in Wuchang direction, and the VIV characteristics of the main span in Hanyang direction are not obvious.

It is noted that in Fig. 3, if 315 mm/s^2 is directly used as a threshold to determine VIV events as Chinese standard (Moreu et al., 2018), although some VIV data can be identified, the entire process of VIV events cannot be fully recognized. This indicates that the method proposed by the standard needs further improvement to more accurately identify the entire process of VIV.

Table 1
Dynamic characteristics of bridge A (Zeng, 2018).

No.	Frequency (Hz)	Damping ratio	Vibration mode
1	0.103	0.0019	First order vertical antisymmetric bending
2	0.135	0.0041	Second order vertical antisymmetric bending
3	0.165	0.0070	First order vertical symmetric bending
4	0.188	0.0061	Third order vertical antisymmetric bending
5	0.208	0.0034	First order asymmetric torsion
6	0.215	0.0039	First order symmetric torsion
7	0.228		Second order asymmetric torsion
8	0.248		Fourth order vertical antisymmetric bending
9	0.268	0.0032	Second order vertical symmetric bending
10	0.310		Third order asymmetric torsion

2.2. Bridge B

Bridge B is located in the East China Sea, connecting two islands. It is an asymmetric steel box girder suspension bridge with a total length of 2713 m. The main span of the bridge is 1650 m. The north and south towers of the suspension bridge are prestressed concrete structures, with a height of 236.5 m. In 2020, the bridge operation and maintenance department upgraded the health monitoring system of Bridge B. Based on the analysis of past monitoring data and bridge vibration characteristics, a new health monitoring system was reinstalled and upgraded, simplifying the sensor layout scheme of the old monitoring system. The new health monitoring system is shown in Fig. 4, and the sampling frequencies of the accelerometers and the anemometer are all 50 Hz.

This study obtained the monitoring dataset of Bridge B from January to September 2021, which includes three acceleration sensors and wind speed and direction sensor data. Due to the large scale of the dataset, it is difficult to identify all VIV events manually. The Chinese standard (Moreu et al., 2018) provides a straightforward classification method: if the RMS of the acceleration sensor data within 10 min exceeds 0.315 m/s^2 and the energy ratio factor is higher than 10, it is identified as a VIV event of the girder. While this method is simple, it has certain drawbacks, such as difficulty in identifying VIV events with smaller acceleration amplitudes. However, it can be easily conducted and find VIV events with large amplitudes in large scale datasets. Therefore, this method was used to find one VIV event in Bridge B and conducted on the datasets of Bridge B in January 2021. The monitoring data for Bridge B in January is shown in Fig. 5, with the top image showing the data from the S2 acceleration sensor on the mid-span. The red dashed line represents the first level warning value of 0.315 m/s^2 VIV acceleration amplitude specified in Chinese standard (Moreu et al., 2018). The middle image shows the wind speed data for January, and the bottom image shows the wind direction data for January. The VIV event identified by Chinese standard is marked by a light red square in Fig. 5.

The 24-h data before and after the abnormal vibration event is plotted in Fig. 6, and the PSD spectrum of the acceleration data beyond the warning range is plotted on the right side. From the time series data, it can be seen that the abnormal vibration event began at 11:09 p.m. on January 21 and ended at approximately 1:50 a.m. on January 22, lasting for nearly 3 h and exhibiting obvious sinusoidal characteristics. From the PSD plots of the acceleration data during this time period, it can be seen that the main frequency of this abnormal vibration is 0.3287 Hz, and this vibration mode is significantly greater than the amplitude of the second main frequency. Therefore, it is determined that the abnormal vibration is VIV event. The fundamental frequencies of the bridge B are identified through finite element model by other researchers and listed in Table 2. It can be concluded that the frequency of the VIV event in

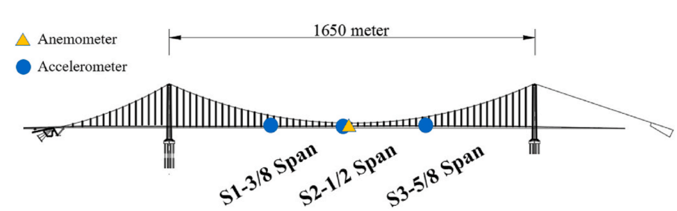


Fig. 4. Partial sensor layout of Bridge B's new health monitoring system.

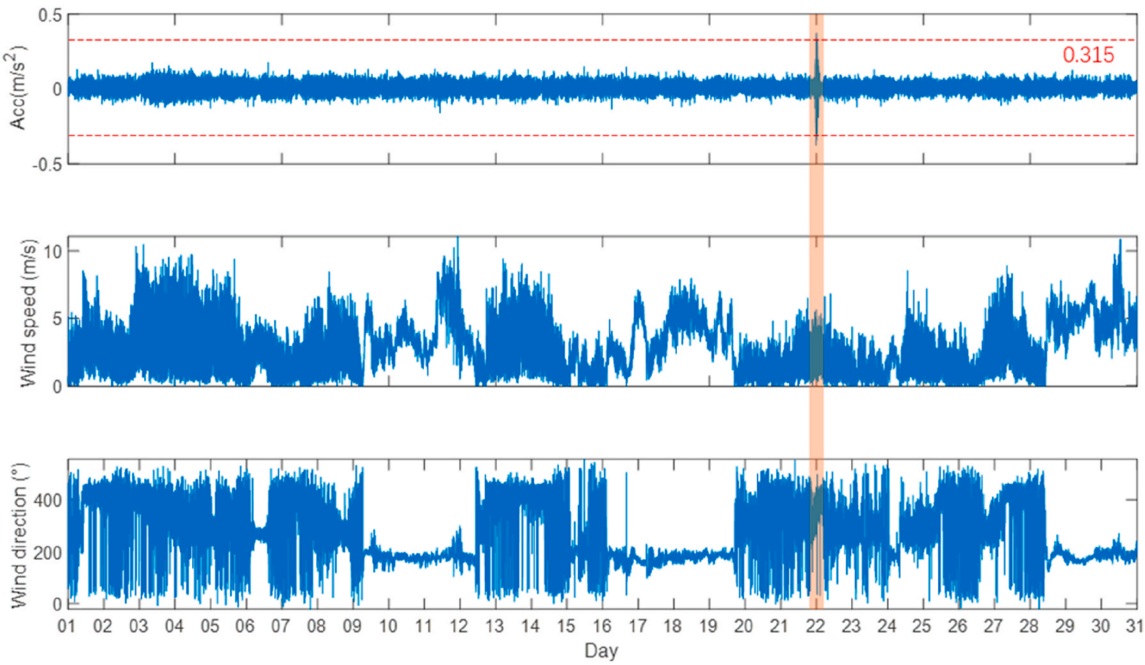


Fig. 5. Bridge B S2 sensor and anemometer monitoring data in January 2021.

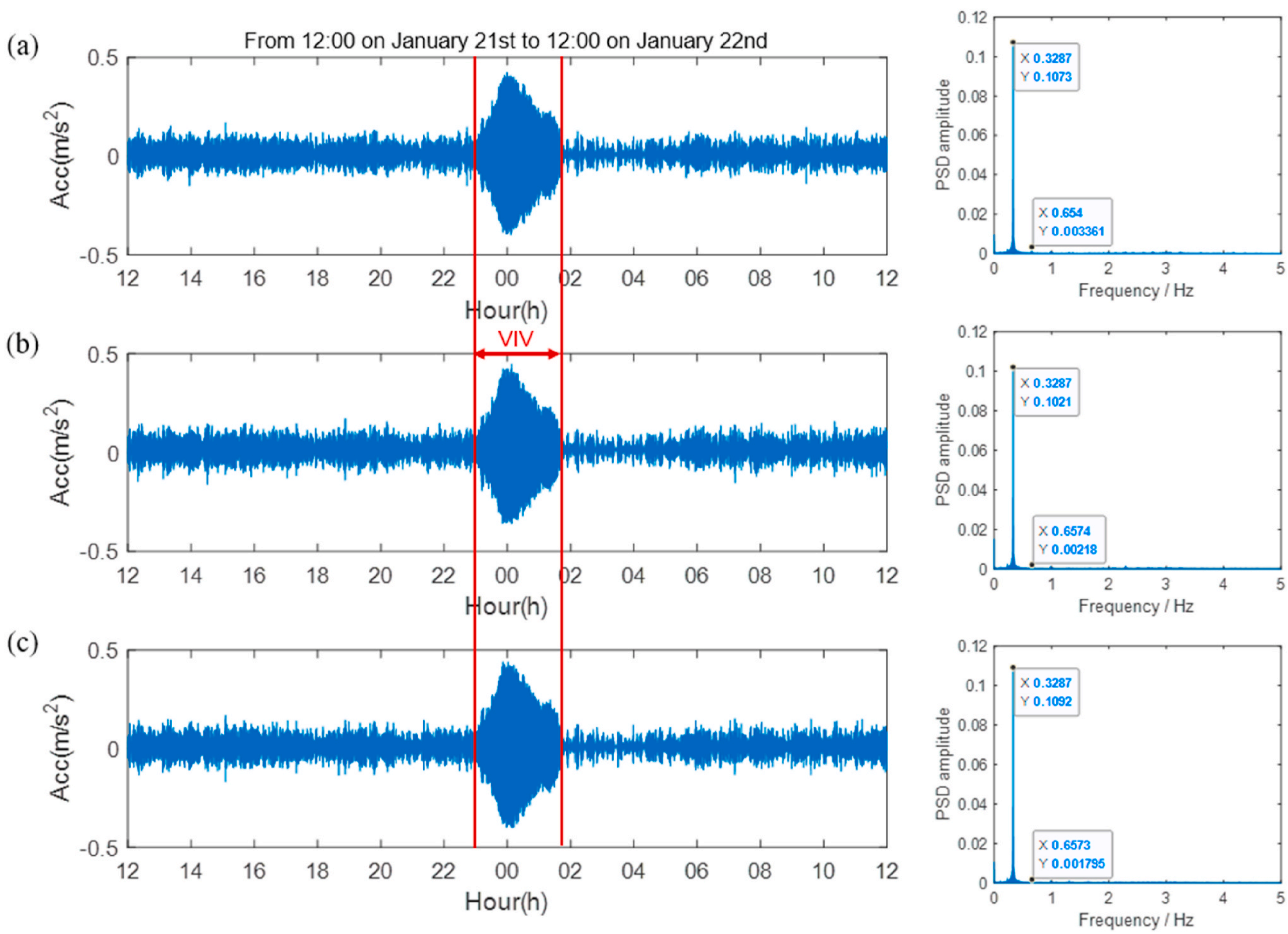


Fig. 6. Abnormal vibration event data and PSD amplitude map of bridge B in January: (a) S1 sensor data and PSD amplitude map; (b) S2 sensor data and PSD amplitude map; (c) S3 sensor data and PSD amplitude map.

Table 2
Dynamic characteristics of bridge B (Li, 2020).

No.	Frequency (Hz)	Vibration mode
1	0.0792	First order vertical antisymmetric bending
2	0.1012	First order vertical symmetric bending
3	0.1129	Second order vertical antisymmetric bending
4	0.1330	Second order vertical symmetric bending
5	0.1790	Third order vertical antisymmetric bending
6	0.1837	Third order vertical symmetric bending
7	0.2295	Fourth order vertical symmetric bending
8	0.2741	Fourth order vertical antisymmetric bending
9	0.3245	Fifth order vertical symmetric bending
10	0.3748	Fifth order vertical antisymmetric bending
11	0.4287	Sixth order vertical symmetric bending
12	0.4827	Sixth order vertical antisymmetric bending

bridge B corresponds to the 9th fundamental frequency 0.325 Hz, which means the mode shape of this VIV event is the fifth order vertical symmetric bending.

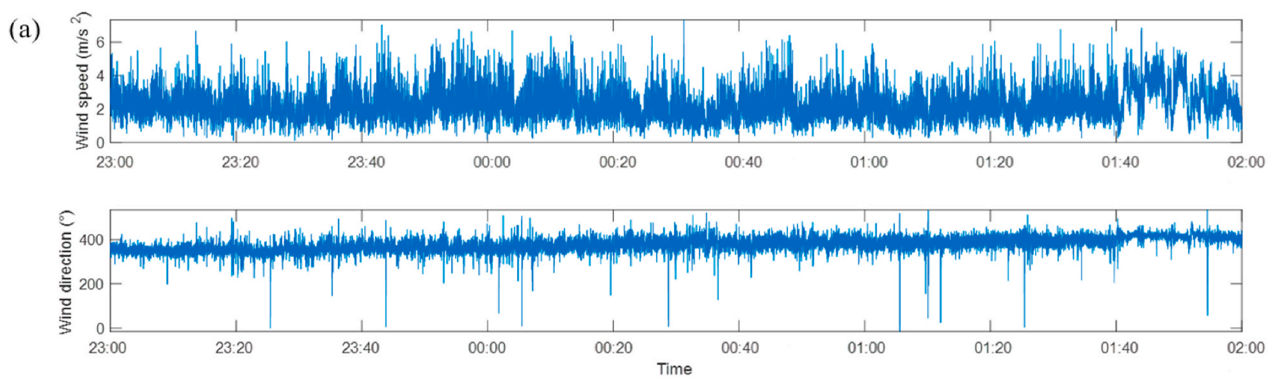
During VIV event, the wind direction remained stable at around 400° and the wind speed remained stable at around 1–5 m/s for about 3 h, as shown in Fig. 7. It can be seen from the wind rose that the wind direction during VIV is mainly vertical to the bridge direction, and the stable wind direction is about 90° from the bridge direction. The wind environment characteristics during this VIV still exhibit stable wind direction and low turbulence. It is worth noting that the wind speed during this VIV was relatively small, and there was no significant increase in wind speed data compared to other time periods. Therefore, it is considered that this VIV is mainly caused by the prolonged stable vertical bridge wind direction.

3. Methodology

Fig. 8 is the schematic diagram of the data-driven adaptive VIV identification method based on domain adaptation methods. Firstly, feature extraction is carried out on the monitoring dataset of the two bridges, the data with NV and VIV label is called source domain data while the monitoring data of the target bridge is called target domain data. Secondly, the feature samples of the source domain with manual NV and VIV labels are aligned with the feature samples of target domain using the unsupervised domain adaptation methods. The feature samples of the source domain and target domain are transformed to adaptive source domain samples and adaptive target domain samples. Lastly, the SVM classifier with Radial Basis Function (RBF) kernel is trained using adaptive source domain samples and adaptive target domain samples. The trained SVM classifier can be used for VIV identification of the target bridge in real-time or in historical monitoring datasets. Herein, the statistical and spectral features in VIV identification are introduced, and the principles of the different domain adaptation methods and SVM classifier are briefly introduced respectively.

3.1. Feature extractions

When VIV occurs on the bridge, the features of the monitoring data (such as acceleration and displacement) will change significantly. Based on the previous research (Arul et al., 2020b; Lim et al., 2022b), the



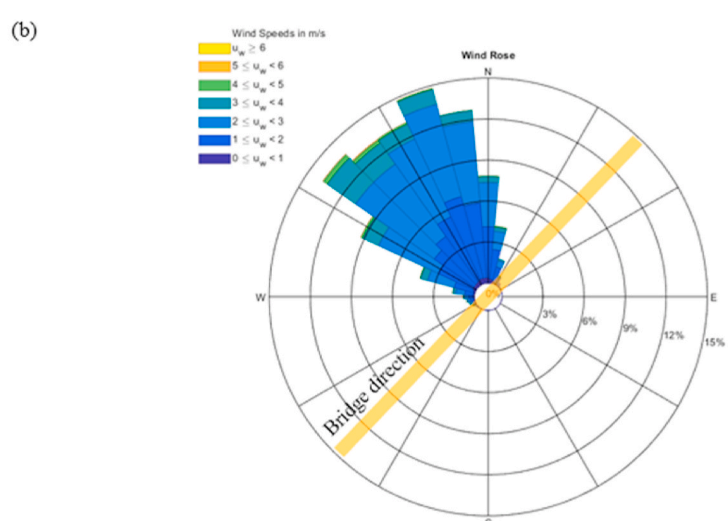


Fig. 7. Wind environment of bridge B VIV phenomenon: (a) time history data of wind speed and direction from 23:00 on January 21st to 2:00 on January 22nd; and (b) wind rose diagram.

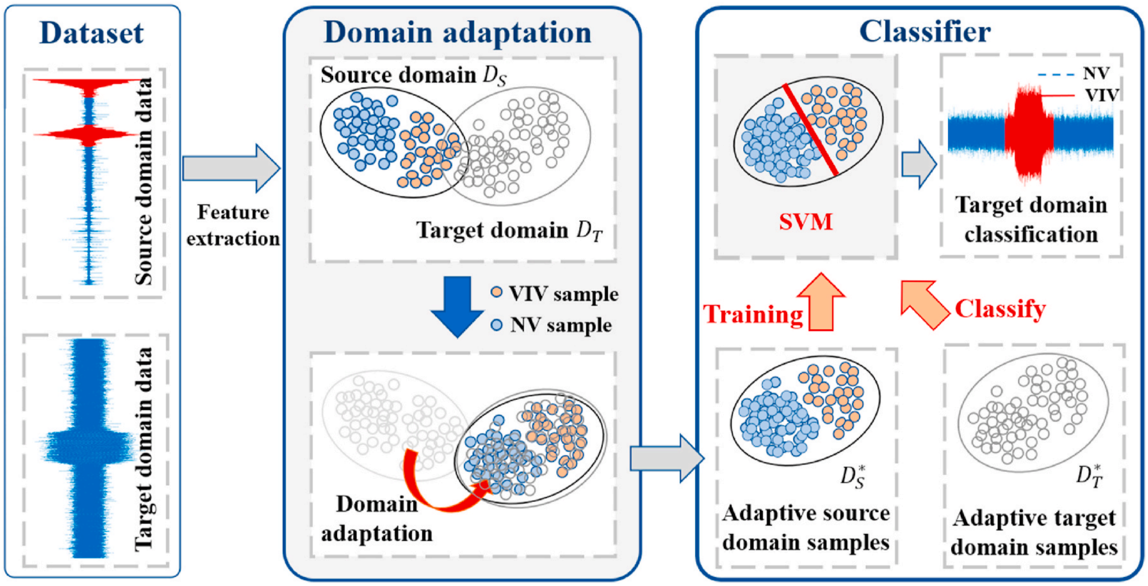


Fig. 8. Schematic diagram of the framework of the adaptive VIV classifier.

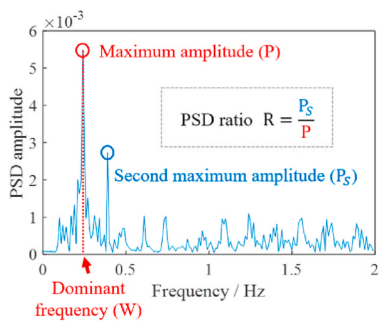


Fig. 9. Definition of the PSD ratio.

following 7 vibration features are selected and extracted from the 10-min-long acceleration sensor signal every minute as shown in Fig. 10. It should be noted that the features are not all used in the proposed VIV recognition method. In Section 4, we will introduce in detail how to use the wrapped filter method to screen and select the optimal feature combination in the VIV recognition task.

3.1.1. Statistical features

There are three statistical features, including root-mean-square (RMS), kurtosis (K), and Hopkins statistic (H). RMS is used to represent the energy magnitude of the vibration signal, K is kurtosis coefficient, which represents the characteristic number of the peak height of the probability density distribution curve at the average value. And H is the randomness of the vibration signal in space. These statistical features are calculated as follow:

$$\mathrm{RMS} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} y_i^2} \tag{1}$$

$$\mathrm{V} = \frac{1}{n}\sum_{i=1}^{n} (y_i - \bar{y})^2 \tag{2}$$

$$\mathrm{H} = \frac{\sum_{i=1}^{m} l_i}{\sum_{i=1}^{m} x_i + \sum_{i=1}^{m} l_i} \tag{3}$$

where n is the number of signal sampling points, y_i is the acceleration or deflection data, and $\bar{y}$ is the mean value of the vibration signal. For H definition, firstly, m points are randomly found from the n points of the data sample $(y_1, y_2, y_3 \ldots\ldots y_m)$. Then for each y_i, a nearest point is found in the sample space, and the minimum distance is x_i, so that a distance vector $(x_1, x_2, x_3 \ldots\ldots x_m)$ is obtained based on the m points. Then randomly generate m points $(s_1, s_2, s_3 \ldots\ldots s_m)$ in the range of possible values of the sample, and find a nearest sample point for each generated point s_i, and calculate this minimum distance to get distance vector $(l_1, l_2, l_3 \ldots\ldots l_m)$.

3.1.2. Spectral features

Three spectral features, the maximum PSD amplitude (P), the PSD ratios (R) and the difference of dominant frequency ($\mathrm{D_w}$) were extracted. Fig. 9 visually defines the P and R, and R is the ratio between the maximum amplitude P and the second PSD maximum amplitude. As for $\mathrm{D_w}$, the 10-min data segment is divided into two 5-min data segments and the difference between the dominant frequency of the first 5-min and the last 5-min is $\mathrm{D_w}$.

3.2. Domain adaptation

Traditional machine learning methods often rely on the foundational assumption of 'independent and identically distributed' data which means the training and test data are sampled independently from a shared distribution (Choi et al., 2020; Janiesch et al., 2021). But not all machine learning tasks can meet this assumption. For example, in this study, when the target bridge does not have sufficient NV and VIV data to construct a VIV identification method for itself, it is necessary to use the monitoring datasets from other bridges as a reference. It is obvious that the feature distributions of the datasets for the two long span bridges are only similar but not the same. Therefore, an important issue is how to "transfer" the known information obtained from one long span bridge with sufficient labeled NV and VIV data to the other. Domain adaptation method is an important part of transfer learning, aiming to map data from different source and target domains into a feature space, making it as close as possible in that space (Wang and Deng, 2018; Wilson and Cook, 2020). Then, the classifier or regression model trained on the source domain in the feature space can be transferred to the target domain, improving the accuracy of the algorithm on the target domain. In our team's previous research, the Transfer Component Analysis (TCA) method has been used in VIV recognition algorithms (Hou et al., 2023). In this research, two more classical domain adaptation methods based on TCA are studied.

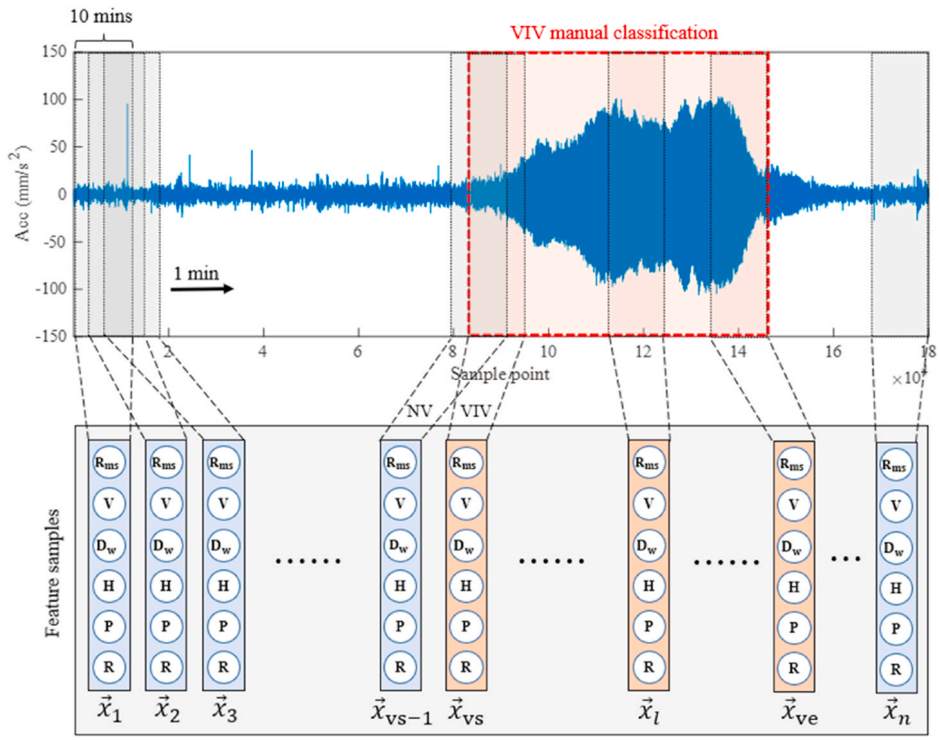


Fig. 10. Feature extraction of vibration signal.

3.2.1. TCA

Transfer Component Analysis (TCA) was initially proposed in 2011 by Yang (Pan et al., 2009) and it is a method of transferring features from source domain to a different but related domain. The following definitions are defined first: the source domain is $\mathbf{X_s} = \{\mathbf{x_{s1}}, \mathbf{x_{s2}}, \mathbf{x_{s3}}, \ldots, \mathbf{x_{sn_s}}\}, \mathbf{x_i} \in \mathbb{R}^d$, the feature sample of source domain is $\mathbf{x_i}$, where $Y_s = \{y_1, y_2, y_3, \ldots, y_{n_s}\}$ and y_i are the corresponding labels of source domain samples, and $\mathrm{n_s}$ is the number of source domain samples. The target domain is $\mathbf{X_t} = \{\mathbf{x_{t1}}, \mathbf{x_{t2}}, \mathbf{x_{t3}}, \ldots, \mathbf{x_{tn_t}}\}, \mathbf{x_i} \in \mathbb{R}^d$, and the target domain has $\mathrm{n_t}$ samples and no labels. TCA begins by assuming the existence of a mapping function ϕ that transfers the source domain $\mathbf{X_s}$ and the target domain $\mathbf{X_t}$ into a new feature space. The objective of this mapping is to establish a similar edge distribution distance between the source and target domains within the designated space $P(\phi(\mathbf{X_s})) \approx P(\phi(\mathbf{X_t}))$, thereby achieving approximate equality in the conditional distribution of both domains $P(\mathbf{Y_s}|\phi(\mathbf{X_s})) \approx P(\mathbf{Y_t}|\phi(\mathbf{X_t}))$.

The primary objective of TCA is to determine an optimal mapping that effectively minimizes the "distance" between the source and target domains within the newly defined feature space. In the realm of distinct distribution comparisons, several methodologies are available for quantifying this "distance," among which is the employment of the KL (Kullback–Leibler) divergence method. However, these methods often necessitate the introduction of additional parameters or intermediate density estimations. In TCA, the researchers utilized an unparameterized distance metric known as the maximum mean discrepancy (MMD) (Bernhard et al., 2007a). This metric was employed to quantitatively assess the distinctiveness between the source and target domains within the framework of the Reproducing Kernel Hilbert Space (RKHS), as exemplified in Equation (4) and $\|\cdot\|_{\mathscr{H}}$ denotes the norm within the RKHS.

$$MMD(\mathbf{X}_s', \mathbf{X}_t') = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(\mathbf{x}_i) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi(\mathbf{x}_j) \right\|_{\mathscr{H}}^2 \tag{4}$$

Generally, ϕ is a function with strong nonlinearity, and directly using the above Equation (4) to optimize ϕ may result in a falling into ϕ poor local optimal solution. A dimensionality reduction-based domain adaptation method called MMDE (Pan et al., 2008) is used in TCA to employ the kernel function $\mathbf{K}$ in place of ϕ. Consequently, the MMD can be represented by Equation (5):

$$MMD(\mathbf{X}_s', \mathbf{X}_t') = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(\mathbf{x}_i) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi(\mathbf{x}_j) \right\|_{\mathscr{H}}^2$$
$$= \frac{1}{n_s^2} \sum_{i=1}^{n_s} \sum_{j=1}^{n_s} k(\mathbf{x_{Si}}, \mathbf{x_{Sj}}) + \frac{1}{n_t^2} \sum_{i=1}^{n_t} \sum_{j=1}^{n_t} k(\mathbf{x_{Ti}}, \mathbf{x_{Tj}}) - \frac{2}{n_s n_t} \sum_{i=1}^{n_s} \sum_{j=1}^{n_t} k(\mathbf{x_{Si}}, \mathbf{x_{Tj}})$$
$$= trace(\mathbf{KL}) \tag{5}$$

$$\mathbf{L}_{ij} = \begin{cases} \dfrac{1}{n_s^2} & \mathbf{x}_i, \mathbf{x}_j \in X_s \\ \dfrac{1}{n_t^2} & \mathbf{x}_i, \mathbf{x}_j \in X_t \\ -\dfrac{1}{n_s n_t} & \text{otherwise} \end{cases} \tag{6}$$

$$\mathbf{K} = \begin{bmatrix} \mathbf{K}_{S,S} & \mathbf{K}_{S,T} \\ \mathbf{K}_{T,S} & \mathbf{K}_{T,T} \end{bmatrix} \tag{7}$$

where $\mathbf{x_{Si}}, \mathbf{x_{Ti}}$ is the i-th sample of the source domain $\mathbf{X}_s'$ and target domain $\mathbf{X}_t'$, and $k(\mathbf{x_{Si}}, \mathbf{x_{Tj}})$ is the kernel function of the i-th sample of $\mathbf{X}_s'$ and j-th sample of $\mathbf{X}_t'$.

In Equation (5), the term "*trace*" represents the sum of the diagonal elements of a matrix. With the problem now reduced to minimizing Equation (5) by solving the kernel matrix $\mathbf{K}$, one approach to tackle this problem is through the use of semi-definite programming (SDP). Nevertheless, the utilization of SDP can impose significant computational demands and resource consumption, rendering it unfeasible for large datasets. In order to mitigate this constraint, TCA decomposes the kernel matrix into $\mathbf{K} = (\mathbf{K}\mathbf{K}^{-1/2})(\mathbf{K}^{-1/2}\mathbf{K})$ and introduces $\widetilde{\mathbf{W}} \in \mathbb{R}^{(n_s+n_t)\times m}$, resulting in a reduction of the kernel matrix to a m dimensional space, where $m\, n_s + n_t$. Substituting into Equation (5) can obtain:

$$MMD(\mathbf{X}_s', \mathbf{X}_t') = trace((\mathbf{K}\mathbf{W}\mathbf{W}^\top\mathbf{K})\mathbf{L}) = trace(\mathbf{W}^\top\mathbf{K}\mathbf{L}\mathbf{K}\mathbf{W}) \tag{8}$$

where $\mathbf{W} = \mathbf{K}^{(-1/2)}\widetilde{\mathbf{W}}$ is transformation matrix. The final optimization objectives of unsupervised TCA are as follows:

$$\begin{aligned} \min_{\mathbf{W}} \quad & trace(\mathbf{W}^\top\mathbf{K}\mathbf{L}\mathbf{K}\mathbf{W}) + \mu \cdot trace(\mathbf{W}^\top\mathbf{W}) \\ \text{s.t.} \quad & \mathbf{W}^\top\mathbf{K}\mathbf{H}\mathbf{K}\mathbf{W} = \mathbf{I}_m \end{aligned} \tag{9}$$

The $trace(\mathbf{W}^\top\mathbf{W})$ is the regularization term of the optimization objective to control the complexity of $\mathbf{W}$, and $\mu > 0$ serves as a trade-off parameter. Similar to kernel Fisher discriminant analysis (Muller et al., 2001), the transformation matrix $\mathbf{W}$ solutions in Equation (9) are the m leading eigenvectors of $(\mathbf{K}\mathbf{L}\mathbf{K} + \mu\mathbf{I})^{-1}\mathbf{K}\mathbf{H}\mathbf{K}$.

3.2.2. JDA

The optimization goal of TCA is to reduce the MMD distance of the edge distribution of features in the hidden space between the source and target domains. On the basis of TCA, Joint Distribution Adaptation (JDA) (Long et al., 2013) incorporates the conditional distribution distance between the source and target domains into its optimization objectives. The mapping matrix $\mathbf{A} \in \mathbb{R}^{d\times k}$ is defined, where d is the feature dimension of the source and target domains which is 6 in this study, and k is the dimension mapped to the new feature space. JDA maps $\mathbf{X} = [\mathbf{X_s} \quad \mathbf{X_t}] \in \mathbb{R}^{d\times(n_s+n_t)}$ to the hidden k-dimensional space matrix $\mathbf{Z}$ by $\mathbf{Z} = \mathbf{A}^\top\mathbf{X}$.

The optimization goal of JDA includes two parts, the first part is the edge distribution distance. Similar to TCA, JDA still uses MMD distance to measure the edge distribution distance between the source and target domains. The edge distribution MMD distance is shown in Equation (9), where $\mathbf{M}_0$ is equal to $\mathbf{L}$ in Equation (6).

$$\left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\mathbf{A}^\top\mathbf{x}_i - \frac{1}{n_t}\sum_{j=1}^{n_t}\mathbf{A}^\top\mathbf{x}_j \right\|^2 = \mathrm{tr}(\mathbf{A}^\top\mathbf{X}\mathbf{M}_0\mathbf{X}^\top\mathbf{A}) \tag{10}$$

The second part is conditional distribution adaptation, with the main goal of adapting the conditional probability distribution of the source and target domains, which means mapping matrix $\mathbf{A}$ also need to make the distance between $P(Y_s|\mathbf{A}^\top\mathbf{X_s})$ and $P(Y_t|\mathbf{A}^\top\mathbf{X_t})$ smaller. JDA still uses MMD distance to measure the conditional distribution distance, as shown below, where n_c and m_c are the number of samples from class c in the source and target domains, respectively. The MMD distance of the conditional distribution is shown in Equation (11). It is worth noting that when measuring the MMD distance of the conditional distribution, the target domain does not have label information, so Equation (11) cannot be used directly. To achieve this, JDA first trains a classifier using source domain data and labels, and applies this classifier directly to target domain data to obtain pseudo labels of the target domain.

$$\sum_{c=1}^{C}\left\| \frac{1}{n_c}\sum_{\mathbf{x}_{s_i}\in\mathscr{D}_s^{(c)}}\mathbf{A}^\top\mathbf{x}_{s_i} - \frac{1}{m_c}\sum_{\mathbf{x}_{t_i}\in\mathscr{D}_t^{(c)}}\mathbf{A}^\top\mathbf{x}_{t_i} \right\|_{\mathscr{H}}^2 = \sum_{c=1}^{C}\mathrm{tr}(\mathbf{A}^\top\mathbf{X}\mathbf{M}_c\mathbf{X}^\top\mathbf{A}) \tag{11}$$

$$(\mathbf{M}_c)_{ij} = \begin{cases} \dfrac{1}{n_c^2} & \mathbf{x}_i, \mathbf{x}_j \in X_s^{(c)} \\ \dfrac{1}{m_c^2} & \mathbf{x}_i, \mathbf{x}_j \in X_t^{(c)} \\ -\dfrac{1}{m_c n_c} & \begin{cases} \mathbf{x}_i \in X_s^{(c)}, \mathbf{x}_j \in X_t^{(c)} \\ \mathbf{x}_i \in X_t^{(c)}, \mathbf{x}_j \in X_s^{(c)} \end{cases} \\ 0 & otherwise \end{cases} \tag{12}$$

Ultimately, the optimization objective for JDA can be obtained as Equation (13), where $\lambda \parallel \mathbf{A}\|_F^2$ is the regularization term. The optimization term with $c = 0$ is the optimization term for edge distribution.

$$\begin{aligned} & \min \sum_{c=0}^{C}\mathrm{tr}(\mathbf{A}^\top\mathbf{X}\mathbf{M}_c\mathbf{X}^\top\mathbf{A}) + \lambda \parallel \mathbf{A}\|_F^2 \\ & \text{s.t.} \quad \mathbf{A}^\top\mathbf{X}\mathbf{H}\mathbf{X}^\top\mathbf{A} = \mathbf{I} \end{aligned} \tag{13}$$

According to Lagrange method and the constrained optimization theory, Lagrange function for Equation (13) can be derived as shown in Equation (14). And $\mathbf{\Phi}$ is the Lagrange multiplier.

$$L = \mathrm{tr}\left(\mathbf{A}^\top\left(\mathbf{X}\sum_{c=0}^{C}\mathbf{X}\mathbf{M}_c\mathbf{X}^\top + \lambda\mathbf{I}\right)\mathbf{A}\right) + \mathrm{tr}((\mathbf{I} - \mathbf{A}^\top\mathbf{X}\mathbf{H}\mathbf{X}^\top\mathbf{A})\mathbf{\Phi}) \tag{14}$$

Setting $\frac{\partial L}{\partial \mathbf{A}} = 0$, the generalized solution for $\mathbf{A}$ can be obtained as follow:

$$\left(\mathbf{X}\sum_{c=0}^{C}\mathbf{M}_c\mathbf{X}^\top + \lambda\mathbf{I}\right)\mathbf{A} = \mathbf{X}\mathbf{H}\mathbf{X}^\top\mathbf{A}\mathbf{\Phi} \tag{15}$$

It is worth noting that unlike TCA, JDA is an iterative algorithm. In the first round, the target domain label in Equation (11) is directly classified by the classifier trained on the source domain data. The initial $\mathbf{A}_0$ can be solved using formula (15). Then, retrain the classifier using $(\mathbf{A}_0{}^\top\mathbf{X_s}, \mathbf{Y_s})$ and classify the target domain data $\mathbf{A}_0{}^\top\mathbf{X_t}$. Then the obtained labels are then used as the target domain label and substituted into Equation (15) to solve for $\mathbf{A}_1$.

3.2.3. BDA

Balanced Distribution Adaptation (BDA) (Wang et al., 2017) has a strong connection with TCA and JDA. JDA uses a mapping matrix $\mathbf{A}$ to reduce both the conditional distribution and edge distribution distances of the source and target domain data. The conditional distribution MMD and edge distribution MMD are directly added to form the final optimization objective of JDA, which is Equation (13). Based on JDA, a balance factor μ is introduced as shown in Equation (14) to balance conditional distribution and edge distribution in the final optimization objective of BDA.

$$\begin{aligned} & \min tr\left(\mathbf{A}^\top\mathbf{X}\left((1-\mu)\mathbf{M}_0 + \sum_{c=0}^{c}\mu\mathbf{M}_c\right)\mathbf{X}^\top\mathbf{A}\right) + \lambda \parallel \mathbf{A}\|_F^2 \\ & \text{s.t.}\mathbf{A}^\top\mathbf{X}\mathbf{H}\mathbf{X}^\top\mathbf{A} = \mathbf{I} \end{aligned} \tag{16}$$

The balance factor in this paper is finely estimated by using A-distance in (Wang et al., 2018; Bernhard et al., 2007b). A-distance is a measure used to quantify the difference between two probability distributions, often applied in domain adaptation and transfer learning. Usually a binary classifier is trained (e.g., a support vector machine or a neural network) to distinguish between the feature representations of the source and target domains. Then A-distance can be defined as follow:

$$d_M = d_A(\mathbf{X_s}, \mathbf{X_t}) = 2(1 - 2\epsilon(h)) \tag{17}$$

$\epsilon(h)$ is the error of the classifier h discriminating the source domain and target domain. For the A-distance between conditional distributions d_c, it can be calculated as $d_c = d_A\left(X_s^{(c)}, X_t^{(c)}\right)$. Eventually, μ can be estimated as:

$$\widehat{\mu} \approx 1 - \frac{d_M}{d_M + \sum_{c=1}^{C} d_c} \tag{18}$$

Similar to JDA, BDA is also an iterative algorithm, with the only difference being that it introduces the balance factor μ.

It is worth noting that for some nonlinear problems, kernel techniques can also be used in JDA and BDA, which is called Kernel JDA (Long et al., 2013) and Kernel BDA (Wang et al., 2017). The kernel tricks in the domain adaptation methods containing two types, linear kernel, Radial Basis Function (RBF) kernel, are studied in this study.

3.3. SVM training

After feature extraction of vibration signals of the source domain and target domain, domain adaptation algorithms are then used to align the source domain samples with the target domain samples. The adaptive source domain is utilized to train a supervised SVM algorithm, and K fold cross validation is used to find the optimal parameters (such as Penalty coefficient C and Gamma G) of SVM. The trained VIV classifier based on SVM can identify the VIV samples in the adaptive target domain. SVM is a classical machine learning algorithm, which have been widely used in engineering field (Lan et al., 2023; Zhang et al., 2018). Since the theory and practice of SVM have been well established, this section does not elaborate on their algorithmic principles. In this study, the SVM classifier is quickly implemented using the Python scikit-learn module (Buitinck et al., 2013), where the kernel functions default to the Radial Basis Function (RBF) kernel.

Algorithm 1. TCA based VIV identification algorithm

Input: the source domain data $\mathbf{X_s}$ and labels Y_s; target domain $\mathbf{X_t}$; kernel type; dimension m
Output: the target domain label Y_t

1. Construct kernel matrix $\mathbf{K}$ and matrix $\mathbf{L}$ from source domain data $\mathbf{X_s}$ and target domain data $\mathbf{X_t}$ by Equation (6) and Equation (7).
2. Eigendecompose the matrix $(\mathbf{KLK} + \mu\mathbf{I})^{-1}\mathbf{KHK}$ and select the m leading eigenvectors to construct the transformation matrix $\mathbf{W}$. Compute the transferred source domain $\mathbf{X_s'}$ and the transferred target domain $\mathbf{X_t'}$ using $\mathbf{W}$.
3. Train the SVM classifier using $\mathbf{X_s'}$, Y_s. Use the SVM classifier to identify VIV data in $\mathbf{X_t'}$ and get the target domain label Y_t.

Algorithm 2. JDA based VIV identification algorithm

Input: the source domain data $\mathbf{X_s}$ and labels Y_s; target domain $\mathbf{X_t}$; kernel type; dimension m
Output: the target domain label Y_t
Begin:
Construct $\mathbf{M}_0$ from source domain data $\mathbf{X_s}$ and target domain data $\mathbf{X_t}$ by Equation (6), and set $\{\mathbf{M}_c := \mathbf{0}\}_{c=1}^{C}$
Repeat:

1. Solve the generalized eigndecomposition problem in Equation (15) and select m leading vector to construct $\mathbf{A}$. Compute the transferred source domain and the transferred target domain by $\mathbf{X_s'} = \mathbf{A}^\top\mathbf{X_s}$ and $\mathbf{X_t'} = \mathbf{A}^\top\mathbf{X_t}$.
2. Train the SVM classifier using $\mathbf{X_s'}$, Y_s. Use the SVM classifier to identify VIV data in $\mathbf{X_t'}$ and get the target domain label Y_t.
3. Construct a new $\mathbf{M}_c$ by Equation (12).

Until convergence
Return: the target domain label Y_t

Algorithm 3. BDA based VIV identification algorithm

Input: the source domain data $\mathbf{X_s}$ and labels Y_s; target domain $\mathbf{X_t}$; kernel type; dimension m
Output: the target domain label Y_t
Begin:
Construct $\mathbf{M}_0$ from source domain data $\mathbf{X_s}$ and target domain data $\mathbf{X_t}$ by Equation (6), and set $\{\mathbf{M}_c := \mathbf{0}\}_{c=1}^{C}$, $\mu = 0$
Repeat:

1. Solve the generalized eigndecomposition problem in Equation (16) and select m leading vector to construct $\mathbf{A}$. Compute the transferred source domain and the transferred target domain by $\mathbf{X_s'} = \mathbf{A}^\top\mathbf{X_s}$ and $\mathbf{X_t'} = \mathbf{A}^\top\mathbf{X_t}$.
2. Train the SVM classifier using $\mathbf{X_s'}$, Y_s. Use the SVM classifier to identify VIV data in $\mathbf{X_t'}$ and get the target domain label Y_t.
3. Compute a new μ by Equation (18).
4. Construct a new $\mathbf{M}_c$ by Equation (12).

Until convergence
Return: the target domain label Y_t

The pseudocode of the algorithm proposed in this article is shown above. The kernel types include Primal, RBF, and Linear. Among them, Primal is the original feature, and the number of dimensions in the high-dimensional space m is equal to the number of features in the input sample. And when the kernel type is set to RBF or Linear, then m is set to 10, slightly higher than the input feature number 6, which can enable the model to achieve strong non-linear ability.

4. Verification

In this section, the two monitoring datasets of long span bridges in Section 2 which both contain NV and VIV data will be used to verify the proposed VIV identification method. The dataset of each of these two bridges can be used as the source domain, while the dataset of the other bridge can be used as the target domain to verify the accuracy of VIV recognition of the proposed methods. It is noted that all experiments were conducted using Python 3.6, and the code can be found at *https://github.com/houseu2000/DA-VIV*.

4.1. Bridge B to bridge A

In this part, assuming a VIV classification algorithm needs to be developed for long span Bridge A, but Bridge A does not have VIV data. Bridge A can be considered as the target bridge, and the dataset of Bridge A is the target domain. However, the January monitoring dataset for another large span Bridge B includes both NV and VIV data. Thus, in the case, the monitoring dataset of bridge B in January can be used as source domain and the VIV identification algorithm is developed for Bridge A without any prior information on Bridge A datasets.

The first scenario: the B1 (S1 sensor of Bridge B) dataset as the source domain, and the AQ (section Q sensor of bridge A) dataset as the target domain is discussed here. Before using the AQ data as the target domain, manual VIV analysis and classification were performed on it as shown in Section 2.1, and the range of VIV was determined. To compare the domain adaptation methods based VIV classification algorithms used in this study, a benchmark model called noDA is proposed, which is trained using source domain data and directly applied to the target domain.

The VIV identification results using different VIV identification methods are shown in Fig. 11, and as discussed in Section 2.1, there are two VIV events in AQ dataset. In Fig. 11, the data displayed within the frame is manually recognized VIV data. It can be seen that compared with noDA algorithm, domain adaptation based VIV identification algorithms can identify more VIV data, and BDA based method can more accurately identify the data of the formation and dissipation stages of VIV events. Among the three domain adaptation based methods, the BDA based method identified the most VIV samples. In this study, four indicators (accuracy, recall, precision and F1 score) are used to evaluate the identification results of the VIV algorithm as shown in Fig. 12.

It is worth noting that none of the domain adaptation methods mentioned above use kernel functions. To investigate the impact of different kernel functions on VIV recognition results, Tabel 1 shows the VIV identification results of AQ dataset using different domain adaptation based VIV identification method with different kernels. From Tables 1 and it can be seen that using kernel functions in the TCA method has a significant impact on the VIV recognition results compared to directly using original features, and only the use of linear kernel functions in the BDA method can improve the accuracy of VIV recognition. The sample features used in this study were extracted using the methods introduced in Section 3.1. This feature extraction method uses known knowledge about vortex induced vibration and extracts features that can clearly distinguish between VIV and NV, making it different from feature extraction methods based on neural networks. Therefore, the VIV classification problem using features in Section 3.1 is not a nonlinear and complex problem, and the use of kernel functions may not necessarily result in better VIV recognition results. In the following part, the domain

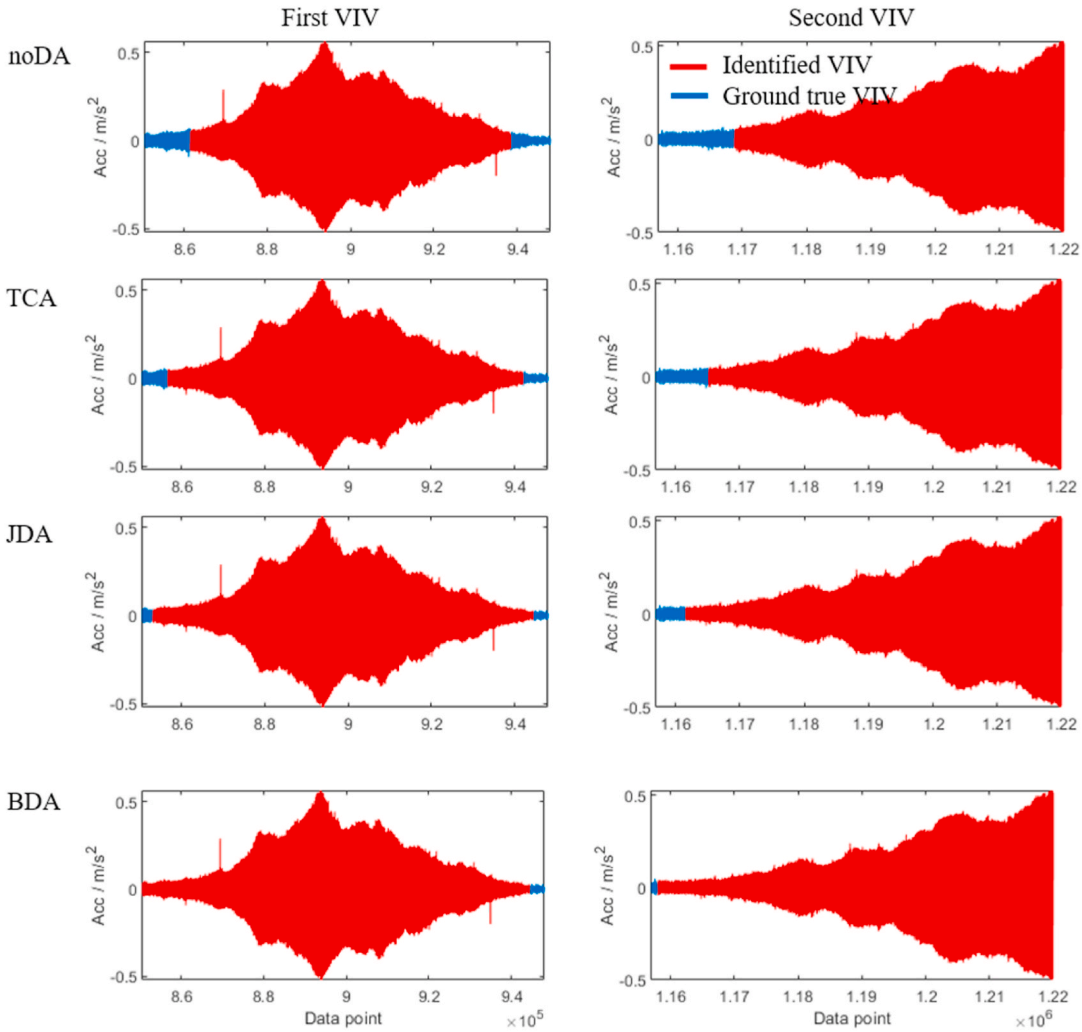


Fig. 11. The VIV identification results of different algorithms in AQ dataset, where B1 dataset used as the source domain.

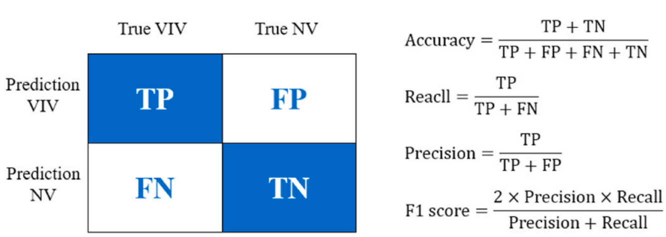


Fig. 12. Evaluation indicators for evaluating VIV identification results.

adaptation based method does not apply to any kernel functions by default.

Furthermore, the features used in the algorithm has a direct impact on the final result. Therefore, in this case, BDA based VIV identification algorithm with different feature combinations are tested for their performance. The results of feature selection are shown in Table 4. It can be seen that regardless of the features used, the BDA based method can achieve better VIV recognition performance than the noDA benchmark model. Although feature combination ‘P Dw RMS R’ has the highest F1 score, it misclassifies 34 NV samples as VIV samples. The classification error of ‘P Dw RMS R V H’ used in this research is the lowest.

Table 3
VIV Identification results of AQ dataset while B1 as the source domain.

Method	Kernel	TN	FN	FP	TP	Accuracy	Recall	Precision	F1 score
noDA	Primal	886	0	33	90	0.97	0.73	1	0.85
TCA	Primal	886	0	23	100	0.98	0.81	1	0.9
	Linear	886	0	52	71	0.95	0.58	1	0.73
	RBF	886	0	52	71	0.95	0.58	1	0.73
JDA	Primal	886	0	15	108	0.99	0.88	1	0.94
	Linear	886	0	15	108	0.99	0.88	1	0.94
	RBF	886	0	21	102	0.98	0.83	1	0.91
BDA	Primal	886	0	9	114	0.99	0.93	1	0.96
	Linear	886	0	7	116	0.99	0.94	1	0.97
	RBF	886	0	11	112	0.99	0.91	1	0.95

Table 4
Feature selection results of the BDA based VIV identification method.

Features	TN	FN	FP	TP	Accuracy	Recall	Precision	F1 score
R V H	872	14	8	115	0.978	0.935	0.984	0.959
P V H	886	0	27	96	0.973	0.780	1.000	0.877
P Dw H	886	0	14	109	0.986	0.886	1.000	0.940
P Dw RMS	830	56	0	123	0.944	1.000	0.937	0.967
RMS R V H	884	2	14	109	0.984	0.886	0.998	0.939
P R V H	881	5	9	114	0.986	0.927	0.994	0.959
P Dw V H	886	0	24	99	0.976	0.805	1.000	0.892
P Dw RMS H	886	0	24	99	0.976	0.805	1.000	0.892
P Dw RMS R	852	34	0	123	0.966	1.000	0.962	**0.980**
Dw RMS R V H	879	7	12	111	0.981	0.902	0.992	0.945
P RMS R V H	885	1	12	111	0.987	0.902	0.999	0.948
P Dw R V H	881	5	12	111	0.983	0.902	0.994	0.946
P Dw RMS V H	886	0	25	98	0.975	0.797	1.000	0.887
P Dw RMS R H	886	0	28	95	0.972	0.772	1.000	0.872
P Dw RMS R V	850	36	0	123	0.964	1.000	0.959	0.979
P Dw RMS R V H	886	0	9	114	**0.991**	0.930	1.000	0.964

In order to verify the robustness of the proposed method, there are other different scenarios to be conducted for validation. Bridge A and Bridge B have sensors at different cross-sectional positions. Here, we can try using sensor data from other sections of Bridge B (such as B1, B2, B3) as the source domain to identify VIV data in section L (AL) and Q (AQ) of Bridge A. From Table 5, it can be seen that the domain adaptation based methods achieved higher recognition accuracy and higher F1 scores compared to the benchmark model in each case. Especially when using the Bridge B1 dataset as the source domain to identify VIV events in the AL dataset, the noDA method's F1 score is only 0.165, only identifying 7 VIV samples. However, the BDA based method's F1 score is 0.861, identifying 59 VIV samples. The same results can also be observed in the identification of VIV in AL using sensors from other cross sections of bridge B, such as B3-AL.

To better demonstrate the effectiveness of BDA, the adapted feature space is illustrated in this section. The case B3-AL, from Table 4 where the BDA algorithm significantly outperformed the benchmark model (noDA), is selected for presentation. As shown in Fig. 13, the adapted feature space of the source domain (B3 data) and the target domain (AL data) is visualized.

The figure illustrates the feature space matching results following domain adaptation across six feature dimensions. To facilitate visualization, the results are presented as scatter plots of feature pairs. The misalignment of parameter H in the scatter plot may seem more pronounced due to the two-dimensional projection, which not reflecting the full alignment across all six dimensions. It is also crucial to consider that the horizontal axis range for parameter H spans from −0.00005 to 0.00025, a scale negligible when compared to the range for parameter R, which extends from −0.2 to 0.4. This substantial difference in scale can visually exaggerate the perceived mismatch for H, making it appear more significant than it is within the context of the entire feature space. Overall, the distribution of the source and target domains in the BDA-calculated feature space is noticeably more similar compared to the original feature space. This visualization underscores the effectiveness of the BDA method in aligning the feature spaces between the source and target domains, thereby enabling more accurate VIV identification across different bridges.

4.2. Bridge A to bridge B

Section 4.1 discusses the situations of using Bridge B as the source domain to separately identify the VIV data in Bridge A monitoring datasets without any prior information on Bridge A datasets. In fact, the proposed methods in this article can identify VIV data of long span bridge's monitoring dataset using a specific bridge dataset (including VIV, NV data, manual labels) as the source domain. In order to verify the applicability and stability of the proposed algorithms in more situations, the cases where the Bridge A dataset is used as the source domain to

Table 5
VIV Identification results in different scenarios, where Bridge B serves as the source domain to identify the VIV of Bridge A.

	Method	TN	FN	FP	TP	Accuracy	Recall	Precision	F1 score
B2-AQ	noDA	886	0	30	93	0.97	0.76	1	0.861
	TCA	886	0	22	101	0.98	0.82	1	0.902
	JDA	886	0	14	109	0.99	0.89	1	0.94
	BDA	886	0	10	113	0.99	0.92	1	**0.958**
B3-AQ	noDA	886	0	33	90	0.967	0.732	1	0.845
	TCA	886	0	23	100	0.977	0.813	1	0.897
	JDA	886	0	8	115	0.992	0.935	1	0.966
	BDA	886	0	7	116	0.993	0.943	1	**0.971**
B1-AL	noDA	931	0	71	7	0.93	0.09	1	0.165
	TCA	931	0	22	56	0.978	0.718	1	0.836
	JDA	931	0	20	58	0.98	0.744	1	0.853
	BDA	931	0	19	59	0.981	0.756	1	**0.861**
B2-AL	noDA	931	0	53	25	0.947	0.321	1	0.485
	TCA	931	0	22	56	0.978	0.718	1	0.836
	JDA	931	0	19	59	0.981	0.756	1	**0.861**
	BDA	931	0	19	59	0.981	0.756	1	**0.861**
B3-AL	noDA	931	0	77	1	0.924	0.013	1	0.025
	TCA	931	0	23	55	0.977	0.705	1	0.827
	JDA	931	0	20	58	0.98	0.744	1	0.853
	BDA	931	0	19	59	0.981	0.756	1	**0.861**

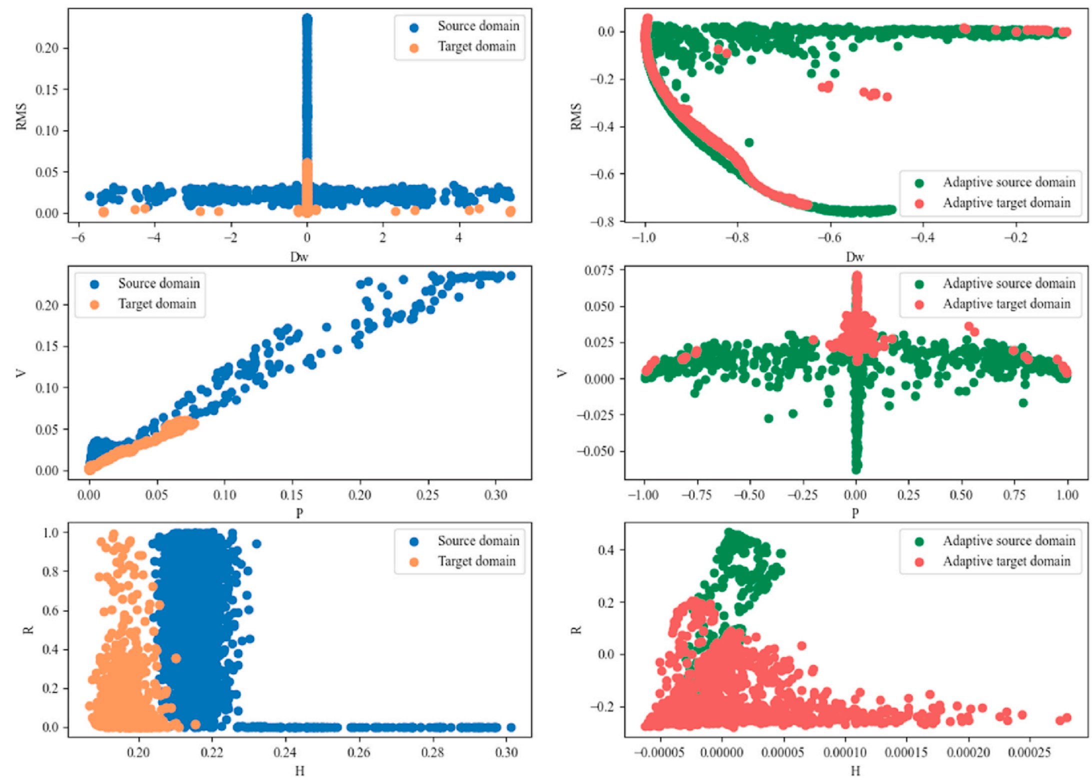


Fig. 13. The original and adapted feature space in the case B3-AL.

identify the VIV data of Bridge B dataset are discussed in this part. The AQ and AL of Bridge A are used as source domains to identify VIV in the B1, B2, and B3 acceleration sensor datasets of Bridge B. The above four indicators (accuracy, recall, precision, F1 score) will still be used to evaluate the VIV identification results of different methods.

The VIV identification results in different scenarios are shown in Table 3. From Table 6, it can be seen that the domain adaptation algorithms (TCA, JDA, BDA) based methods significantly identified more VIV samples in the Bridge B dataset compared with noDA benchmark model, and BDA based method identified more VIV samples compared with TCA and JDA based methods in each case. The conclusion drawn in this section is basically consistent with the conclusion in Section 4.1. The BDA based method has the highest F1 score, identifies the most VIV samples, and does not mistakenly identify NV samples as VIV samples, which means it is the optimal VIV identification method based on domain adaptation method.

Table 6
VIV identification results in different scenarios, where Bridge A serves as the source domain to identify the VIV of Bridge B.

	Method	TN	FN	FP	TP	Accuracy	Recall	Precision	F1 score
AQ-B1	noDA	2710	0	46	114	0.98	0.71	1	0.832
	TCA	2710	0	19	141	0.99	0.88	1	0.937
	JDA	2710	0	19	141	0.99	0.88	1	0.937
	BDA	2710	0	18	142	0.99	0.89	1	**0.940**
AQ-B2	noDA	2710	0	49	111	0.98	0.69	1	0.819
	TCA	2710	0	20	140	0.99	0.88	1	0.933
	JDA	2710	0	20	140	0.99	0.88	1	0.933
	BDA	2710	0	19	141	0.99	0.88	1	**0.937**
AQ-B3	noDA	2710	0	44	116	0.985	0.725	1	0.841
	TCA	2710	0	19	141	0.993	0.881	1	0.937
	JDA	2710	0	19	141	0.993	0.881	1	0.937
	BDA	2710	0	17	143	0.994	0.894	1	**0.944**
AL-B1	noDA	2710	0	14	146	0.995	0.913	1	0.954
	TCA	2710	0	10	150	0.997	0.938	1	0.968
	JDA	2710	0	11	149	0.996	0.931	1	0.964
	BDA	2710	0	8	152	0.997	0.950	1	**0.974**
AL-B2	noDA	2710	0	14	146	0.995	0.913	1	0.954
	TCA	2710	0	10	150	0.997	0.938	1	0.968
	JDA	2710	0	12	148	0.996	0.925	1	0.961
	BDA	2710	0	8	152	0.997	0.950	1	**0.974**
AL-B3	noDA	2710	0	12	148	0.996	0.925	1	0.961
	TCA	2710	0	10	150	0.997	0.938	1	0.968
	JDA	2710	0	11	149	0.996	0.931	1	0.964
	BDA	2710	0	8	152	0.997	0.950	1	**0.974**

5. Application

Bridge B has the monitoring dataset from January to September 2021 which includes three acceleration sensors and wind speed and direction sensor data. Due to the large scale of the dataset, manual analysis was only conducted on the data of Bridge B in January 2021 in Section 2.2. In the validation section, the BDA based VIV recognition method identified the most VIV samples and achieved the highest F1 score. Thus, in this section, the BDA based VIV identification method is directly applied on the monitoring datasets of Bridge B from February to September without any priori information of Bridge B to identify VIV events in these eight months. This section takes the AQ datasets as source domain and the S2 section sensor, located at the mid span position of Bridge B, as the target domain, so all VIV events of the girder in Bridge B with positive symmetry of vibration modes can be identified.

5.1. VIV identification in B2 eight-months monitoring datasets

After extracting feature samples from the B2 dataset according to the method in section 3.1, the number of vibration samples per month is approximately 40000. Directly using one month's samples as the target domain can result in excessive computational complexity and slow operation in BDA algorithm. A month's samples are divided into multiple groups, with each group consisting of 3000 samples, and then use each group as the target domain input for BDA based VIV identification algorithm. The 23 VIV events ultimately identified in the B2 dataset are shown in Table 7. Table 7 also shows the duration, RMS, dominant frequency, and average wind speed and direction of each VIV event.

In fact, the VIV events identified by the BDA based algorithm in Table 4 are not necessarily actual VIV events, as the algorithm may mistakenly identify NV samples as VIV samples. Further manual analysis and judgment are required to determine whether the VIV events identified by the algorithm are true or not. Taking the recognition results in March as an example, as shown in Fig. 14. In March, a total of three VIV events were identified, all of which exhibited high RMS values and low PSD ratio R, and the main frequency of the vibration was around 0.328 Hz, which showed the same characteristics as the VIV event in January in Section 2. Therefore, these three VIV events conform to the VIV characteristics and are considered VIV events.

The manual validation and evaluation are performed on each identified VIV events in Table 4, as done for the identified VIV events in March. Due to article length limitations, it is not possible to display the time series and frequency diagrams of each VIV event here. It is found that the recognition results listed in Table 4 all have obvious VIV features and are all VIV events, which means BDA based VIV identification method correctly identified 23 VIV events in B2 datasets.

Table 7
VIV events identified in B2 datasets from February to September.

No.	Month	Duration (min)	RMS (m/s^2)	Frequency (Hz)	Wind speed (m/s)	Wind direction (°)
1	2	197	0.092	0.329	2.499	356.492
2	2	120	0.059	0.328	2.111	344.433
3	3	42	0.052	0.327	2.415	333.499
4	3	111	0.122	0.327	2.388	326.932
5	3	35	0.058	0.327	7.357	165.282
6	5	36	0.051	0.327	7.119	179.012
7	5	16	0.040	0.327	7.722	181.126
8	5	97	0.122	0.326	7.785	176.146
9	5	91	0.171	0.326	7.834	178.794
10	5	16	0.021	0.326	7.725	177.424
11	5	22	0.048	**0.229**	1.776	361.753
12	5	39	0.049	0.327	1.935	366.976
13	5	20	0.050	0.327	2.261	380.824
14	5	46	0.062	0.326	1.799	379.526
15	5	85	0.072	0.326	2.063	373.977
16	7	62	0.140	0.326	7.147	181.569
17	7	27	0.046	0.326	7.608	180.044
18	7	21	0.045	0.327	8.252	164.419
19	7	16	0.036	0.327	8.288	169.366
20	7	14	0.033	0.326	7.361	178.335
21	8	34	0.055	**0.229**	1.755	334.859
22	8	93	0.128	0.326	7.166	177.222
23	9	19	0.047	**0.437**	2.840	364.717

5.2. VIV events analysis

In the previous section, the method proposed in this article was used to identify 23 VIV events in the B2 sensor monitoring dataset without prior information on bridge B. Each VIV event of a bridge has different wind environmental and vibration characteristics. This section mainly analyzes the relationship between the wind speed, wind direction characteristics and vibration characteristics in these 23 VIV events.

The amplitude and dominant frequency are the two most important characteristics of VIV events. Scatter plot of the main frequency and RMS of the acceleration of the identified 23 VIV events in Table 4 is shown in Fig. 15. It can be seen that there are three different main frequency VIV events identified in the 23 vortex vibration events, most of which are VIV events with a frequency of 0.327 Hz.The VIV events with 0.327 Hz frequency have significantly higher RMS values than VIV events with other frequencies. VIV events at other main frequencies occurred only three times: 0.437 Hz occurred once, and 0.229 Hz occurred twice. The number of VIV events at these two frequencies (0.437 Hz and 0.229 Hz) is relatively small, which does not necessarily imply that VIV events at these frequencies will not exhibit large vibration amplitudes in the future.

As is well known, the occurrence and frequency of VIV on the main beam of a long span bridge are closely related to the wind environmental on the bridge (Li et al., 2011b, 2014). This article identified 23 VIV events of girder in Bridge B, and the scatter plots of wind speed and direction characteristics at the time of these VIV events are shown in Fig. 15. It can be seen from Fig. 16 that the wind speed and direction characteristics are clearly distributed in two regions, which means, when the wind speed is about 2.2 m/s and wind direction is about 370°, or when the wind speed is about 7.5 m/s and wind direction is about 175°, Bridge B is prone to occur VIV. The one with the lower wind speed in these two wind environments is referred to as the class A wind environment, and the other is referred to as the class B wind environment. The conclusion can be drawn that the directions of these two wind environments are almost perpendicular to the axis of the Bridge B, but the wind speed conditions are significantly different under different wind directions. When the wind direction is 370°, the wind speed condition for VIV to occur is approximately 2.2 m/s, but when the wind direction is 175°, the wind speed condition for VIV to occur is approximately 7.5 m/s.

By plotting the wind speed and direction scatter points of VIV events with different main frequencies in different colors, it can be seen that VIV events with a main frequency of 0.327 Hz can occur in both A and B wind environments. The VIV events with main frequencies of 0.229 Hz and 0.437 Hz both occurred in Class A wind environments. The number of VIV events with these two frequencies (0.437 Hz and 0.229 Hz) is relatively small, so it does not necessarily mean that Class B wind environments will not cause VIV events with these two frequencies.

The scatter plot of the average wind speed and RMS data of the VIV events identified in Table 4 is shown in Fig. 17, it can be seen that even in class A wind environments with low wind speeds, the VIV event with large amplitude will occur. It is worth noting that in the class B wind environment, Bridge B experienced multiple VIV events with smaller amplitudes. Although the highest amplitude of VIV events occurred in class B wind environments with higher wind speeds, there was no significant trend that higher wind speed leads to higher VIV amplitude. Further research is needed on the correlation between wind speed and the RMS value of VIV events through more monitoring data.

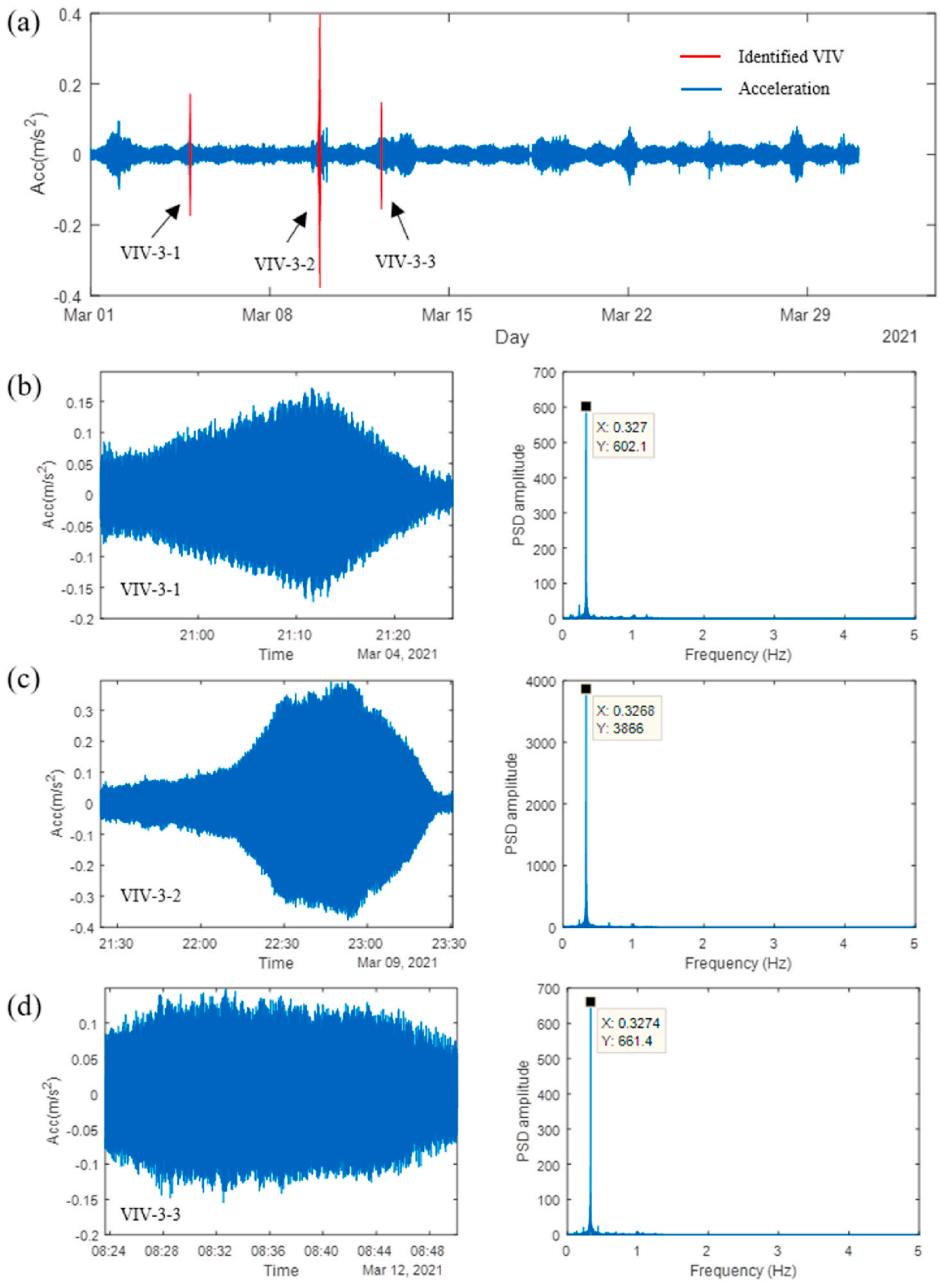


Fig. 14. VIV events identified using the BDA based algorithm in March in the B2 datasets: (a) the identified three VIV events; (b) the time-domain signals and frequency-domain analysis of VIV 3-1; (c) the time-domain signals and frequency-domain analysis of VIV 3-2; (d) the time-domain signals and frequency-domain analysis of VIV 3-3.

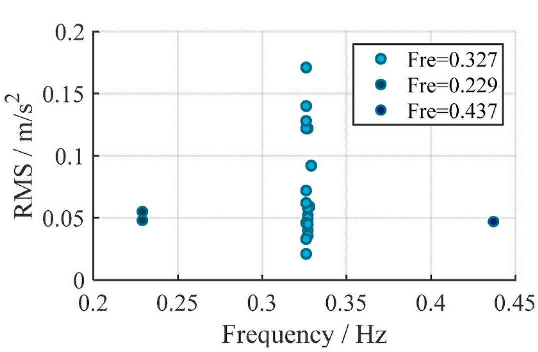


Fig. 15. Scatter plot of RMS of the acceleration and frequency of 23 VIV events.

6. Conclusion

This paper proposes a domain adaptation based vortex induced vibration (VIV) identification algorithm of long span bridge that does not require any prior information of the target bridge. The algorithm is validated on two large span bridges and applied to an 8-month monitoring dataset of one such bridge. The following conclusions are drawn:

a) The domain adaptation based VIV recognition methods are significantly superior to the benchmark model which is trained directly on the source domain data. Among these methods, the Balanced Distribution Adaptation (BDA) based VIV identification method outperforms other domain adaptation based VIV identification methods. In the validation section, the BDA based method achieved the highest

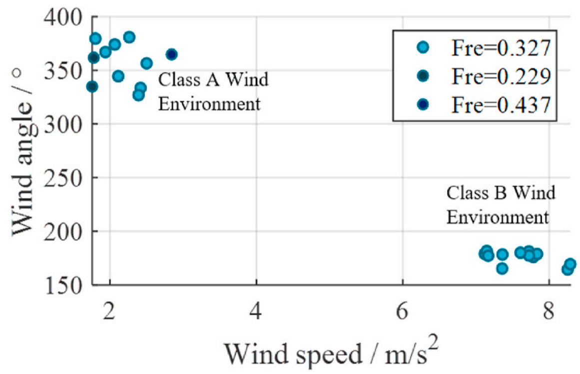


Fig. 16. Scatter plot of wind speed and direction of 23 VIV events.

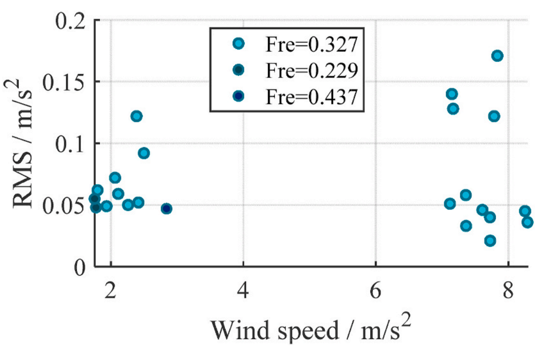


Fig. 17. Scatter plot of wind speed and RMS of 23 VIV events.

F1 score and recognized the most VIV samples across various scenarios.

b) In the dataset from the mid span cross-section sensor of Bridge B, using the Q cross-section sensor data from Bridge A as the source domain, 23 events were identified as VIV events of girder in Bridge B using the BDA based VIV identification method. After manual verification, all identified events match the characteristics of VIV and are confirmed to be VIV events.

c) Through the wind environment analysis of the 23 identified VIV events, it was found that Bridge B is prone to occur VIV when exposed to a wind speed of around 2.2 m/s from a wind direction of around 370°, or a wind environment with a wind speed of around 7.5 m/s from a wind direction of around 175°.

In future research, the method proposed in this article will be applied to analyze more bridge monitoring data, obtaining more VIV samples of large-span bridges to study the relationship between VIV vibration characteristics and wind environment characteristics. And the deep learning-based domain adaptation methods, like Domain Adversarial Neural Networks (DANN) (Ganin et al., 2017) should be investigated in future work.

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Chunfeng Wan: Writing – original draft, Supervision, Data curation, Conceptualization. **Jiale Hou:** Writing – original draft, Methodology, Formal analysis. **Guangcai Zhang:** Writing – review & editing, Project administration, Investigation, Formal analysis. **Shuai Gao:** Validation, Methodology. **Youliang Ding:** Writing – review & editing, Supervision, Funding acquisition. **Sugong Cao:** Supervision, Resources, Project administration, Data curation. **Hao Hu:** Writing – review & editing, Data curation. **Songtao Xue:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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